ZZ-H-461c
September 18, 1964
SUPERSEDING
Fed. Spec. ZZ-H-461b
April 24, 1947
(See 6.4)

FEDERAL SPECIFICATION

HOSE, RUBBER, GAS (ACETYLENE-HYDROGEN, AIR AND OXYGEN)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers acetylene-hydrogen, air and oxygen hoses for cutting, welding and other oxygen and fuel gas uses.

1.2 Classification.

1.2.1 *Types and styles.* Gas hoses covered by this specification shall be of the following types and styles, as specified, (see 6.2):

Type I—Wrapped
- Style A—Single line
- Style B—Dual line

Type II—Braided
- Style A—Single line
- Style B—Dual line

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-271—Paper, Wrapping, Waterproofed Kraft.
ZZ-H-499—Hose, Pneumatic, Wrapped.
CCC-T-191—Textile Test Methods.

Federal Standard:

Fed. Std. No. 601—Rubber; Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-H-775—Hose, Rubber or Fabric (Including Tubing); and Fittings, Nozzles, and Strainers, Packaging of.

Military Standards:

MS15011—Fittings, Hose, Welding and Cutting, Oxygen and Acetylene.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-166—Visual Inspection Guide for Rubber Hose.

MIL-STD-147 — Rubber Compositions Vulcanized, General Purpose Solid (Symbols and Tests).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues in effect on date of invitation for bids shall apply.

Navy Department Specification:

General Specification for Inspection of Material.

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The hose furnished under this specification shall consist of a rubber tube, a layer or layers of cotton or synthetic fabric or braid reinforcement, and a rubber cover.

3.2 Tube and cover. The tube and cover shall be of natural rubber, synthetic rubber or mixtures thereof, which will be referred to herein as rubber.

3.3 Couplings. Couplings and ferrules shall be wrought or cast with chemical composition as shown in table I, and shall comply with the requirements of MS15011.

TABLE I—*Chemical requirements*

Elements	Forgings	Castings
	Percent	*Percent*
Copper, minimum	58.0	76.0
Tin, maximum	1.50	1.50
Iron, maximum	0.20	0.20
Lead, maximum	3.75	3.75
Zinc	remainder	remainder

3.4 Ferrules. Ferrules shall be a minimum of one inch in length, except that the 1/8-, 3/16-, and 1/4-inch-diameter hose ferrule shall be 11/16-inch minimum length, with the inside diameter comparable to the outside diameter of the hose. The ferrules shall be crimped in a manner to insure that the couplings will not become loosened or detached from the hose ends when tested in accordance with the hydrostatic test pressures specified in table II.

3.5 Construction.

3.5.1 ***Type I, wrapped.*** **Wrapped hose shall consist of a rubber inner tube, plies of cotton or synthetic fabric reinforcement and a rubber cover.**

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TABLE II—*Physical requirements*

Size, inside diameter ..... inch	1/8	3/16	1/4	5/16	3/8
Tolerance, inside diameter (plus or minus) .. inch	0.023	0.023	0.031	0.031	0.031
Size, outside diameter, type I, wrapped ..... inch	—	—	0.562	0.625	0.719
Tolerance, outside diameter, type I, wrapped (plus or minus) ..... inch	—	—	0.031	0.031	0.031
Size, outside diameter, type II braided ..... inch	0.469	0.531	0.656	0.719	0.781
Tolerance, outside diameter, type II, braided (plus or minus) ..... inch	0.031	0.031	0.031	0.031	0.031
Thickness, minimum, inch:					
Tube	0.031	0.031	0.040	0.040	0.040
Cover	.042	.042	.045	.045	.045
Tensile strength, minimum, p.s.i.:					
Tube	—	700	700	700	700
Cover		1800	1800	1800	1800
Ultimate elongation, minimum percent:					
Tube	—	200	200	200	200
Cover	—	350	350	350	350
Friction, minimum, lbs.:					
Between tube and plies	—	5	6	7	7
Between cover and plies	—	5	5	5	5
Hydrostatic test pressure, minimum, p.s.i.:					
Coupling and imperfection test	250	250	250	250	250
Burst test	600	600	700	700	700

3.5.2 *Type II, braided.* Braided hose shall consist of a rubber inner tube, braided cotton or synthetic reinforcements, and a rubber cover.

3.5.3 *Inner tube.* The inner tube shall be a seamless, continuous rubber extrusion extending the full length of the hose.

3.5.4 *Style* **A.** Single line hose shall consist of a single line, constructed in accordance with this specification, (see 6.2).

3.5.5 *Style B.* Dual line hose shall consist of two lines, constructed in accordance with this specification (see 6.2), and connected continuously along one side so that hoses are one integral unit. The minimum pull to separate one-inch length shall be 10 pounds for the 1/8-inch size, 15 pounds for the 3/16-inch size, 25 pounds for the 1/4-inch size, 30 pounds for the 5/16-inch size, and 35 pounds for the 3/8-inch size. The lines must separate without exposing the braid.

3.5.5.1 *Split lead.* Definite cut lengths and coupled lengths of dual line hose shall be separated for approximately 12 inches on one end and 30 inches on the other end, and at this juncture a figure 8 shaped brass ferrule shall be crimped to prevent further separation.

3.6 Reinforcement.

3.6.1 *Fabric.* Cotton or synthetic fiber fabric shall be evenly and firmly woven and as free from unsightly defects, dirt, knots, lumps, and irregularities of twist as is consistent with the best manufacturing practices.

3.6.2 *Type I hose.* The layers of fabric shall be applied on a bias of 45 degrees with edges lapped (not sewed) at least one-half inch. The layer or layers of fabric shall be thoroughly impregnated with a rubber compound which shall cause the plies to adhere firmly to each other and to the tube and cover, (see 6.2).

3.6.3 *Type II hose.* The ply or plies of reinforcements shall be braided over the tube. Unless otherwise specified, the number of plies of reinforcement shall be two or more except for the 1/8 and 3/16 inch sizes where one ply shall be used. The braided plies shall be thoroughly impregnated with a rubber

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compound which shall cause the plies to adhere firmly to each other and to the tube and cover. There shall be a distinct layer of rubber compound between the plies, (see 6.2).

3.7 Dimensions.

3.7.1 *Sizes.* Hose covered by this specification shall be of the sizes shown in table II. The braided type only shall be of the 1/8- and 3/16-inch sizes.

3.7.2 *Diameter tolerances.* Hose diameter tolerances shall be as shown in table II.

3.7.3 *Lengths.* Length of hose shall be as specified, (see 6.2). The length shall be the actual length of the hose exclusive of fittings (couplings). A tolerance of plus or minus 2 percent in the specific length shall be permitted as shown in table II.

3.7.4 *Color.* On both dual and single line hose, the cover shall be red for acetylene-hydrogen hose, green for oxygen hose and black for air hose, (see 6.2).

3.7.5 *Marking identification.* The hose shall be branded with the manufacturer's name and trademark, quarter and year of manufacture, and the words "ACET-HYD" for acetylene-hydrogen hose, "OXYGEN" for oxygen hose, "AIR" for air hose, and W.P.—150 lbs. Braided hose less than 25 feet long shall have one brand per length. Hose of 25 feet or more in length shall be branded at intervals of 25 feet. Wrapped hose shall have two markings per 50-foot length, one located approximately 4 feet from each end.

3.7.6 *Flexibility.* Five-sixteenth-inch hose shall deflect from the horizontal not less than the values shown in table III when subjected to the tests specified in 4.4.4.5. Deflections of 3/8-inch, 1/4-inch, and 3/16-inch hose shall be not less than 80, 125, and 150 percent, respectively, of the deflections shown for 5/16-inch hose.

TABLE III—*Flexibility requirements of 5/16-inch hose*

Load	Deflection	
	at 80° ± 5°F.	at −20° ± 2°F.
Ounces	*Inches*	*Inches*
4	5/8	1/4
8	1-1/4	1/2
12	2-1/4	1
15	3	1-1/2

3.7.7 *Low temperature brittleness.* Specimens from the cover and tube shall not fracture or show cracks when tested in accordance with 4.4.4.5.

3.7.8 *Resistance to light.* The tensile strength and ultimate elongation of the cover after accelerated light exposure as described in 4.4.4.1, shall not decrease more than 25 percent from the tensile strength and elongation of the cover before light exposure.

3.7.9 *Physical properties after aging.* The tensile strength and ultimate elongation of the tube and cover, after being subjected to the accelerated aging test specified in 4.4.4.4, shall not show a decrease from the tensile strength and ultimate elongation before aging of over 25 percent unless the test specimens are of necessity less than 0.05-inch thick, in which event the decreases shall not be over 35 percent.

3.8 Workmanship. The workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification

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TABLE IV—*Sampling for visual and dimensional inspection hydrostatic coupling test AQL (approximate)—1.5% defective for major defects and for hydrostatic coupling test 6.5 percent for major and minor defects combined.*

Number of lengths in lot	Number of lengths of hose in sample	AQL—1.5 percent		AQL—6.5 percent	
		Acceptance number (defective)	Rejection number (defective)	Acceptance number (defective)	Rejection number (defective)
15 and under	7	0	1	1	2
16 to 40	10	0	1	1	2
41 to 110	15	0	1	2	3
111 to 300	25	1	2	3	4
301 to 500	35	1	2	5	6
501 and over	50	2	3	6	7

where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot inspection.

4.2.1 *Inspection lot.* For purposes of sampling a lot shall consist of not more than 5000 feet of hose of the same type, style, and size offered for delivery at one time.

4.2.2 *Sampling for inspection and hydrostatic coupling tests.* From each lot of hose with coupling attached, the Government inspector shall select a random sample in accordance with table IV to be subjected to inspection as specified in 4.3 and the hydrostatic coupling test specified in 4.4.2.

4.2.3 *Sampling for hydrostatic burst test and physical tests.* From lots of hose in 10-foot lengths and over, the Government inspector shall select a random sample of lengths in accordance with table V. From each length of hose so selected a 3-foot section shall be cut off for the hydrostatic burst test after which the physical test shall be made on the same sample. The contractor shall refit couplings to said lengths, (see 6.2).

4.2.4 *Lengths under 10 feet.* For lengths under 10 feet the contractor shall furnish extra lengths for test purposes. The number of extra lengths shall be in accordance with table V, (see 6.2).

4.3 Visual and dimensional inspection. Each of the sample lengths of hose selected in accordance with 4.2.2 shall be inspected to verify conformance to all of the requirements of this specification regarding both hose and couplings which do not involve tests. The classification of defects applying to this particular hose, if available, and MIL-STD-166 shall be used to determine and evaluate defects through visual Government inspection.

4.3.1 *Fifty-foot lengths.* Any length of hose containing one or more dimensional defects or more than four minor defects shall be rejected and counted against the AQL for major defects. Any length of hose con-

TABLE V—*Sampling for hydrostatic burst and physical test*

Number of lengths of hose in inspection lot	Number of lengths of hose in sample	Acceptance number (defective)	Rejection number (defective)
15 and under	2	0	1
16 to 40	3	0	1
41 to 110	5	0	1
111 to 300	7	0	1
301 to 500	10	0	1
501 and over	15	0	1

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taining four or less minor defects and no major defects shall be counted a minor defect against the AQL for total defects, but shall not be rejected.

4.3.2 *Lengths other than 50 feet.* For other lengths the number of minor defects which shall cause a length of hose to be rejected and counted against the AQL for major defects shall be proportional to the length.

4.3.3 If, in any sample, the number of defective lengths as defined above exceeds the acceptance number for either AQL for that sample the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 *Preparation of dual line hose.* Dual line hose shall be separated into single lines before conducting any test except that of the web.

4.4.2 *Hydrostatic coupling tests.* Each sample length of hose selected in accordance with 4.2.3 shall be subjected to a hydrostatic coupling test at the pressure specified in table II to verify conformance with this specification. Any length of hose found to be nonconforming shall be rejected and if the number of rejected lengths exceeds the acceptance number specified in table IV the lot shall be rejected.

4.4.3 *Hydrostatic burst tests.* Each of the sample sections selected in accordance with 4.2.4 shall be subjected to a hydrostatic burst test at the pressures specified in table II. If any sample section fails this test, the lot shall be rejected.

4.4.4 *Physical tests.* Each of the sample sections of hose selected in accordance with table IV, after performance of the hydrostatic burst test, shall be subjected to the tests specified in 4.4.4.1 to 4.4.4.5.2 inclusive. If any section is found to be nonconforming on one or more of these tests, the lot shall be rejected.

4.4.4.1 Procedures not specified herein shall be conducted in accordance with the methods specified in Fed. Std. No. 601.

4.4.4.2 *Tensile strength and elongation tests.* Specimens for tensile strength and elongation tests shall conform to type B specimen shown in Fed. Std. No. 601. Each specimen shall be tested to verify conformance with the applicable requirement.

4.4.4.3 *Web test.* Compliance with 3.5.5 shall be determined as follows: with a joined sample 1 inch long, a rod shall be inserted through the bore of each of the two hoses and the rod suitably fixed in a tensile machine. The jaws shall move at a rate of 1 inch per minute. The load required to separate the two hoses shall be measured in pounds.

4.4.4.4 *Aging test.* Compliance with 3.7.9 shall be determined after specimens have been subjected to oxygen at a temperature of 158° ± 2°F. and a pressure of 300 ± 10 p.s.i. for a period of 48 hours.

4.4.4.5 *Flexibility test.* Compliance with 3.7.6 shall be determined as follows:

4.4.4.5.1 *At 80 degrees ± 5 degrees F.* The specimen shall be mounted in the test chamber of apparatus similar to that illustrated in figures 1 and 2. Prior to mounting in the test chamber, the specimen shall be conditioned for 5 hours at 80° ± 5°F. The specimen shall be slipped over a 5/16-inch-diameter rod fixed horizontally in the chamber so that deflection can be measured over a 6-inch length. A chain or wire shall be attached to the free end of the hose, extending over the ballbearing pulleys at the top of the unit to the loading pan. A scale shall be mounted horizontally beside the chain or wire at the top of the unit to measure the distance of travel of the chain or wire with increments of load. Weights shall be added to the pan at intervals of 30 seconds and deflection measured 5 seconds after application of the load at 80° ± 5°F.

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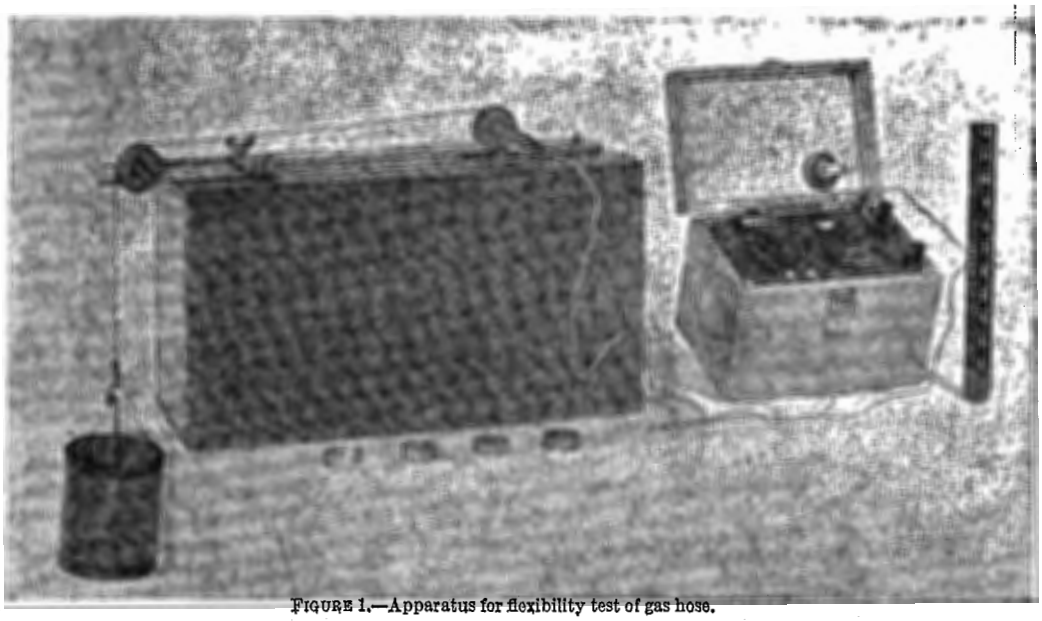


FIGURE 1.—Apparatus for flexibility test of gas hose.

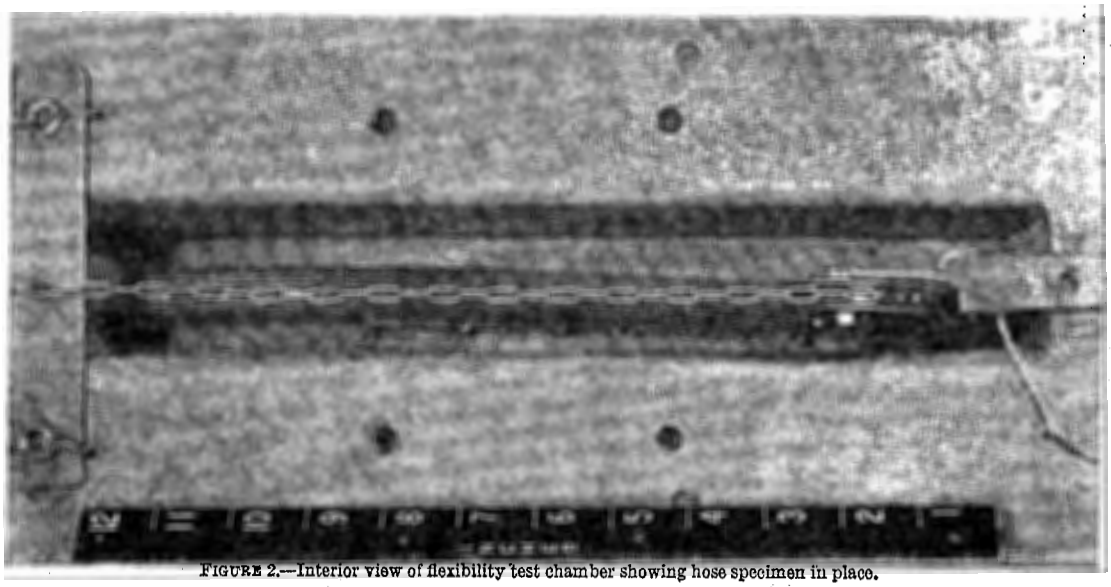


FIGURE 2.—Interior view of flexibility test chamber showing hose specimen in place.

Deflection shall be taken as the average of four measurements made at 90° rotation of the hose.

4.4.4.5.2 *At -20 degrees ± 2 degrees F.* Using the same apparatus as used for the test at 80°F., the specimen shall be mounted in the same manner as described in 4.4.4.5.1. Prior to mounting in the test chamber, the specimen shall be conditioned for 5 hours -20° ± 2°F. The test chamber shall be filled with ethyl alcohol, cooled to -20°F, to within approximately 1-1/2 inch of the top. Dry ice shall be added to maintain the temperature of -20° ± 2°F. The temperature of the alcohol shall be measured with a copper-constantan thermo-couple, a potentiometer, or a suitable thermometer. The tank shall be equipped with a stirrer or circulating pump. Weights shall be added to the pan at intervals of 30 seconds and deflection measured 5 seconds after application of the load -20° ± 2°F.

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4.5 Rejected lots. Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming lengths of hose. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the hose and fittings shall be examined to determine compliance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The hose and fittings shall be preserved, packaged, and packed for each level of protection and marked in accordance with MIL-H-775, (see 6.2).

5.2 Marking.

5.2.1 *Coils and reels.* Each coil or reel shall be marked with the name of the article, the type, the length, and the number of lengths therein, as specified in the contract or order under which shipment is made, and the name of the manufacturer.

5.2.2 *Shipping containers or bales.* Each bale or shipping container shall be tagged or labeled with waterproof material. Each tag or label shall contain the name of the article, the type, the length, and the number of lengths therein, as specified in the contract or order under which shipment is made, and the quarter and year of manufacture, the name of the contractor, the number of the contract or order, and gross weight.

5.2.3 In addition to the foregoing any special marking required by the contract or order shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The intended use for hose covered by this specification is as follows:

Inside diameter

Inch	Intended use
3/16	Types AA and AB torches (Spec. 41T25) for sheet metal work, aircraft, etc.
1/4	Types C and D torches (MIL-T-18068) and type E torches (MIL-T-16183) for medium working pressures and medium gas volumes.
5/16 and 3/8	For high working pressures and high gas volume operations, such as machine and manual production, cutting of steel shapes, and underwater cutting.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, size, color of hose, and purpose for which intended, (see 1.2 and 3.7.4).

(*c*) Length of braided hose when lengths longer than 50 feet are required, (see 3.7.3).

(*d*) Length of wrapped hose, when lengths less than 50 feet are required, (see 3.7.3).

(*e*) That the lengths of hose from which 3-foot sections are taken for hydrostatic burst test will be accepted as full lengths provided the 3-foot sections meet the hydrostatic burst test, and the lengths from which they are taken meet the other requirements of this specification, (see 4.2.3).

(*f*) That where hose is furnished in less than 10-foot lengths, extra lengths shall be furnished by the contractor for test purposes without expense to the Government, (see 4.2.4).

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(*g*) Level of preservation and packaging and level of packing required, (see 5.1).

(*h*) Special marking required, (see 5.2).

6.3 Hose should be ordered in lengths not exceeding 50 feet, unless longer lengths are required for a specific purpose, (see 3.7.3). In such cases, braided hose may be obtained in longer lengths as specified, not to exceed 500 feet.

6.4 **Supersession data.** This specification includes the requirements of MIL-H-1142A, dated April 19, 1961.

CUSTODIANS:

Army—Corps of Engineers

Navy—Bureau of Ships

Air Force—69

Other Interest:

Army—TSIGO

Navy—ASY

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